



Protocol Adaptations for a Sovereign Agentic Web in Solid Ecosystems*

(* position paper)

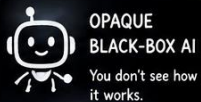
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This presentation includes images generated using GenAI tools.

TODAY

Centralized AI Platform

BIG TECH CLOUD



USERS DATA MODELS INSIGHTS CONTROL



Data locked in silos



No transparency



Platform controls you



Vendor lock-in



Weak privacy & sovereignty

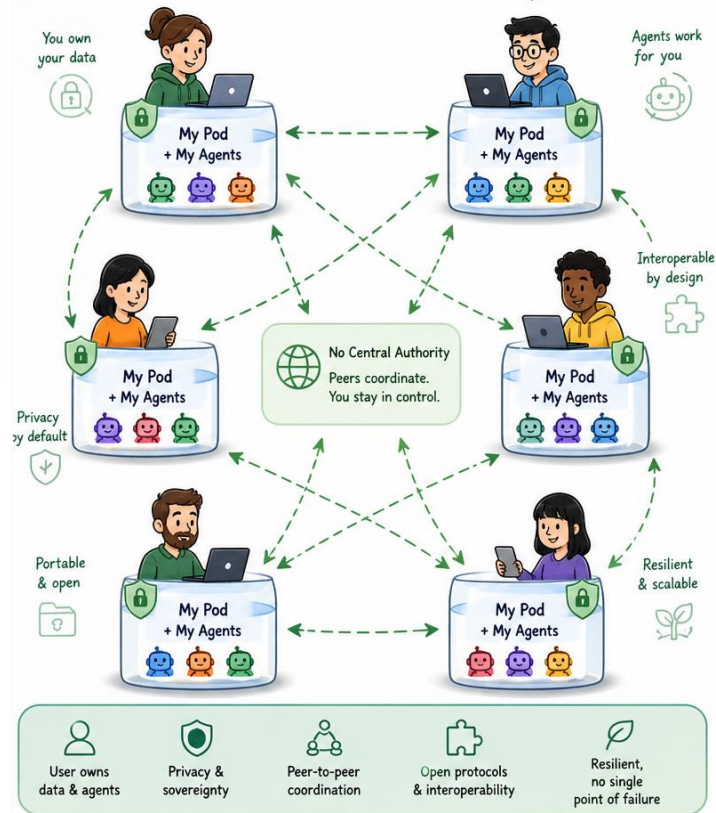
Today's AI assistants are mostly hosted in corporate clouds.

What kind of protocol adaptations are needed to make our AI assistants **sovereign citizens?**



SOVEREIGN AGENTIC WEB

Decentralized • User-Owned • Interoperable



Value Proposition for Solid

Solid already provides:

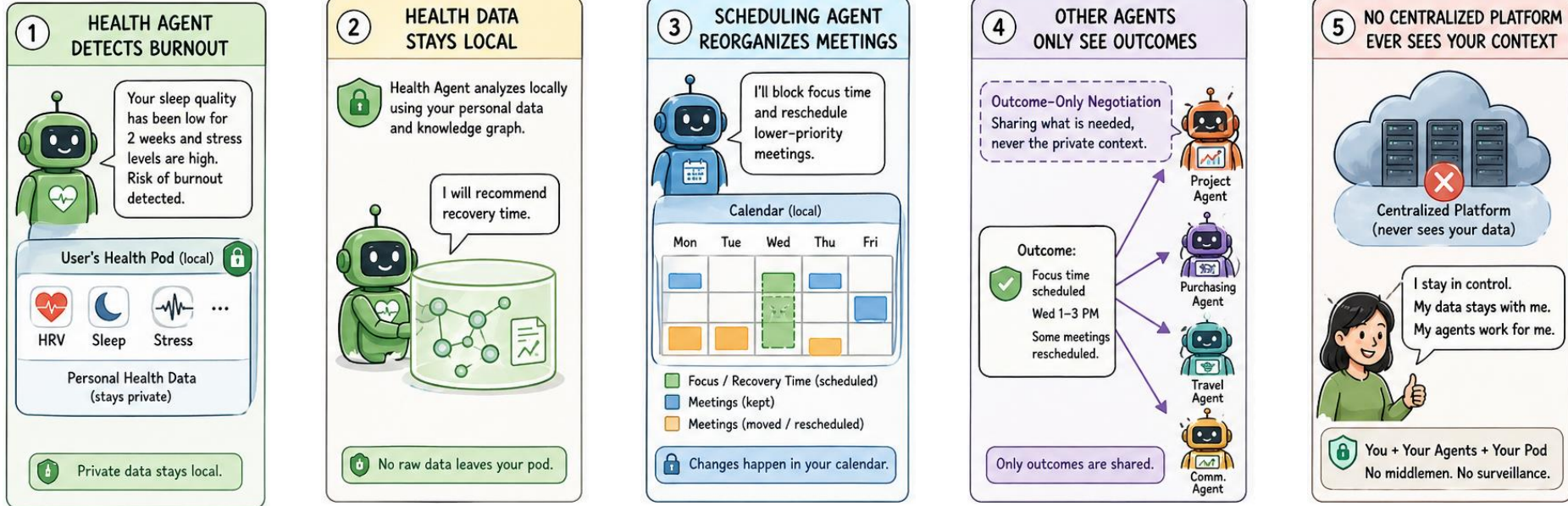
- Decentralized Identity
- Web-native Access Control
- Linked Data
- Interoperable Storage

But Solid pods are still treated as **passive data stores**.

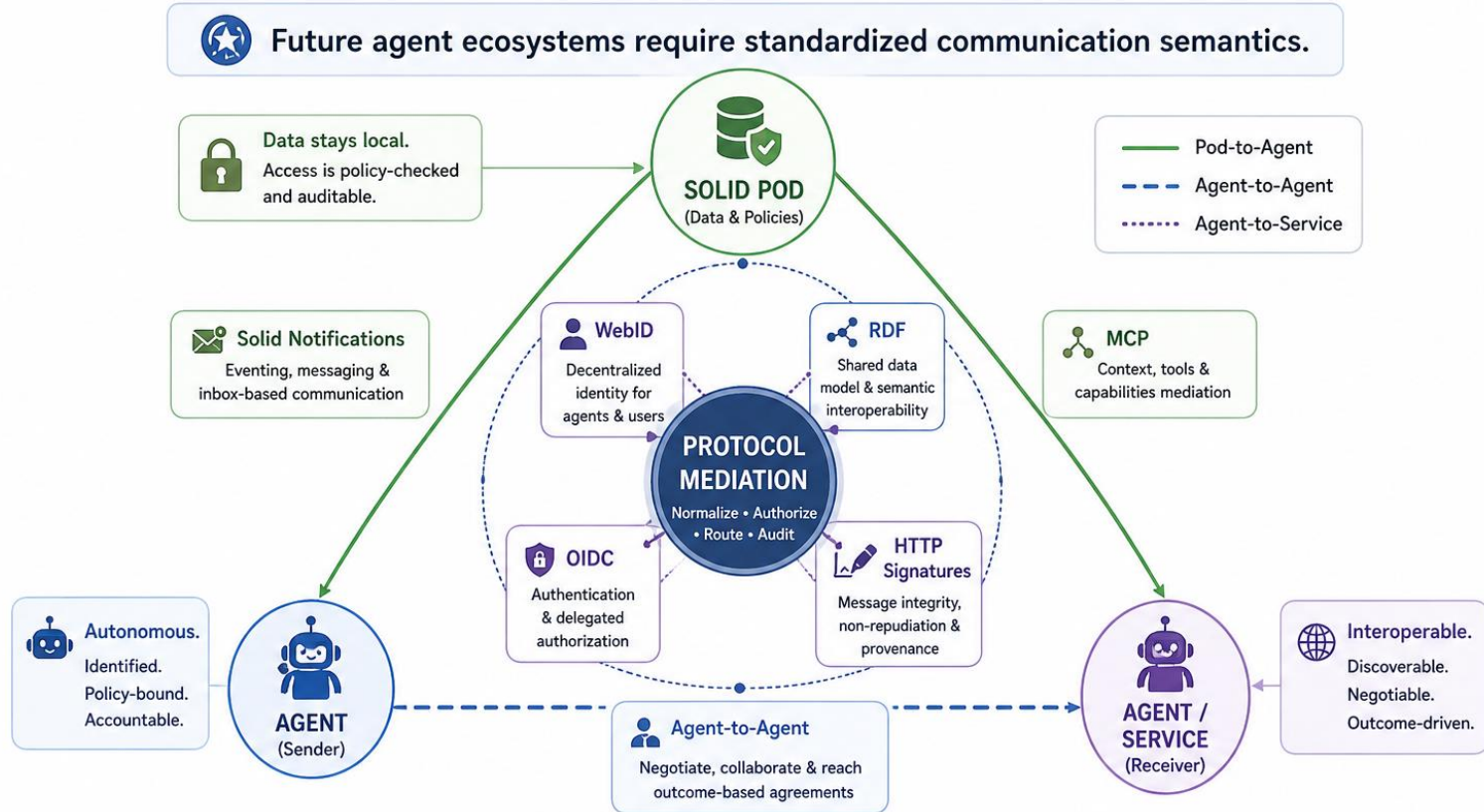
We argue that Solid could facilitate an ecosystem where:

- Pods host:
 - Semantic memory/ Context graphs
 - Workflows
 - Policy mediation
- Agents as Web Principals with:
 - WebIDs
 - Revocable Capabilities
 - Auditability
- Federation for Coordination

Motivating Use Case

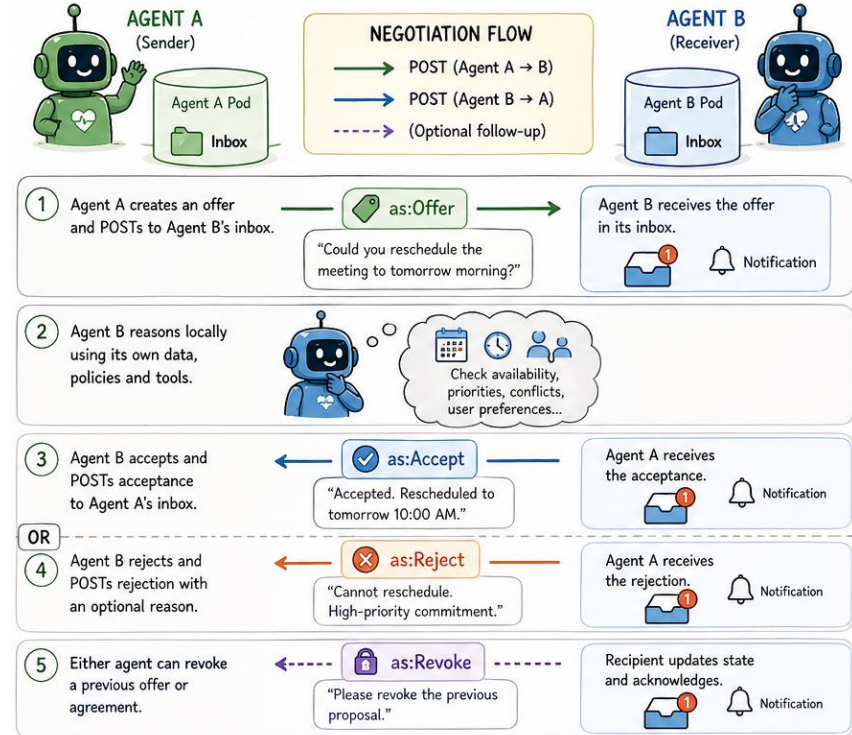
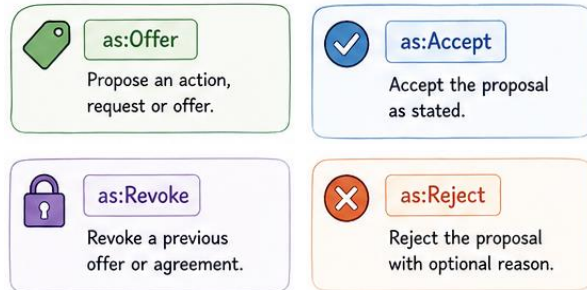


Protocol-Centric Communication Model



Extending Solid Notifications for Negotiation

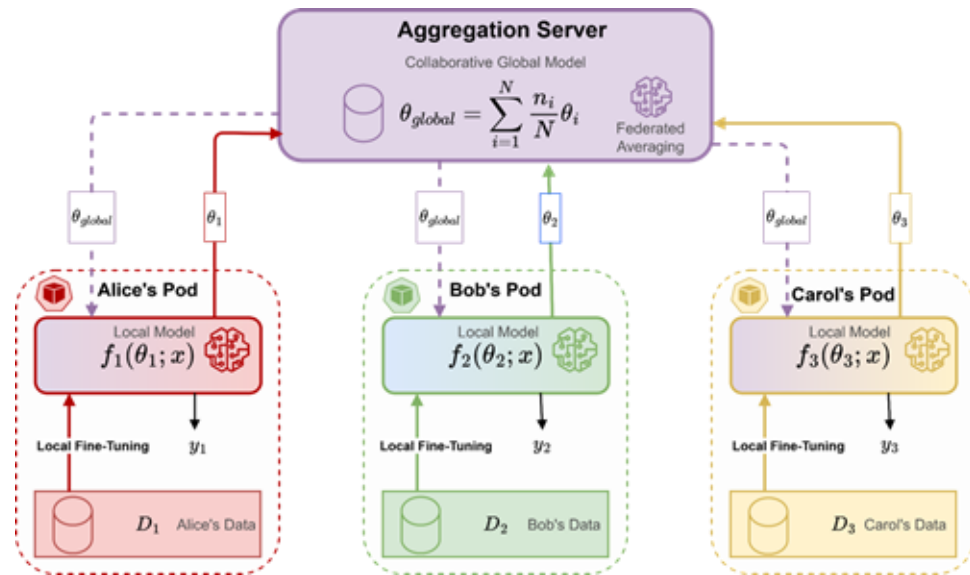
- Existing Solid notifications:
 - deliver events.
- But agents need:
 - offers,
 - acceptance,
 - revocation,
 - negotiation,
 - asynchronous coordination.



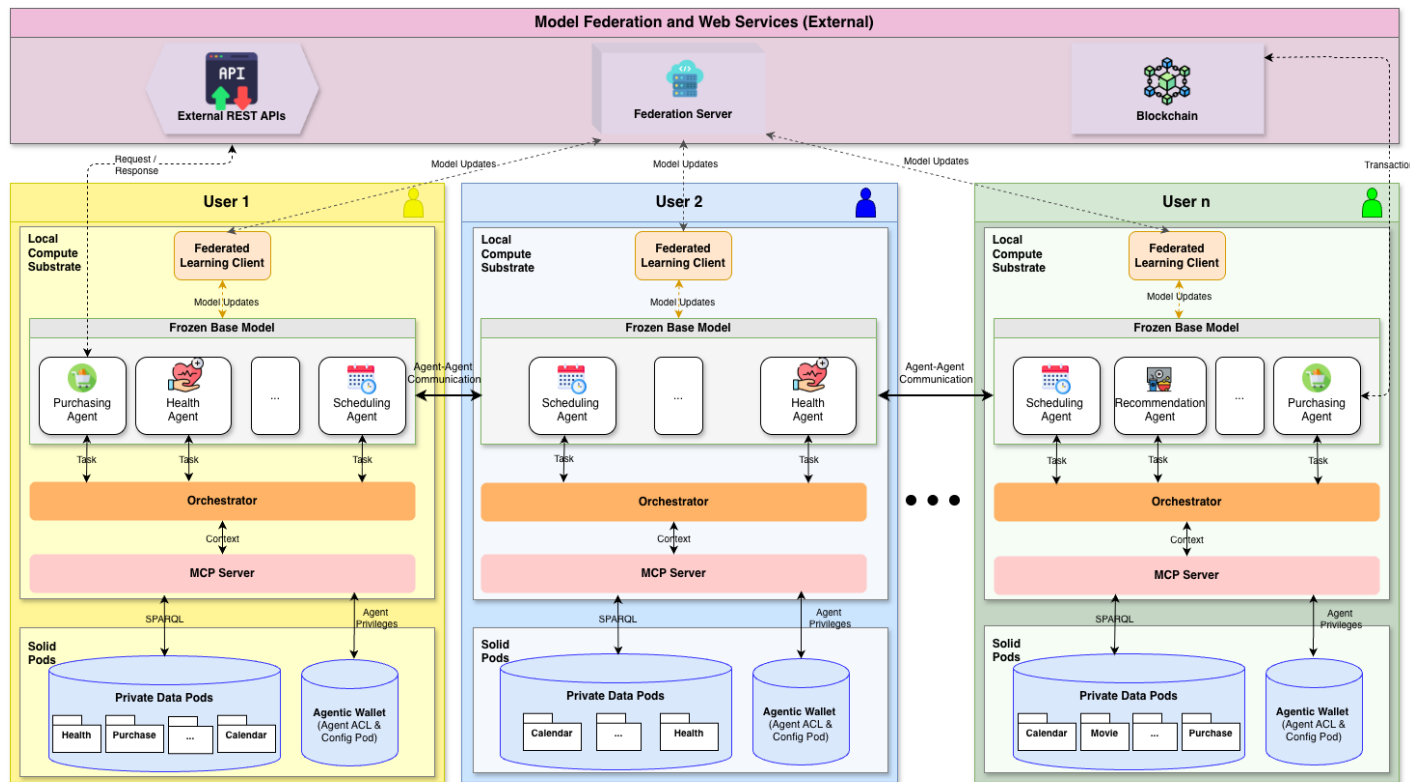
Privacy Preserving Federated Learning

Unifies four core technologies:

- ✓ **Personal Knowledge Graphs:** On-device data sovereignty & user interpretability.
- ✓ **Lightweight LLMs (Qwen3):** On-device reasoning and context-aware generation.
- ✓ **Federated Learning:** Decentralized collaborative intelligence without raw data sharing.
- ✓ **Solid Pods:** Data Storage and retrieval mechanism.

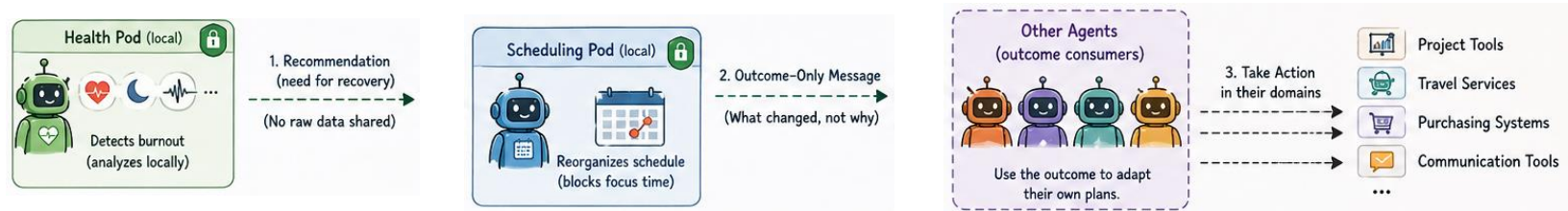


Proposed Architecture



Summary

- This position paper proposes to **extend Solid beyond storage** to facilitate agentic AI.
- Proposes lightweight interoperable agent semantics and adopt open Web principles for agentic interactions to enable decentralized AI coordination.
- Use case:



Web → Social Web → Decentralized Web → Agentic Web

Questions?

Email us with questions at spadef@rpi.edu or senevo@rpi.edu